



Arithmetic with extensional fuzzy numbers

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ABSTRACT

Fuzzy numbers are commonly used to represent imprecise data and are typically modelled using fuzzy sets. However, arithmetic operations on fuzzy numbers are not as straightforward as those on crisp values. The classical approach based on Zadeh's extension principle often leads to three major issues: (1) high computational cost due to iterative procedures, (2) lack of closure under operations (e.g., multiplication of triangular fuzzy numbers does not yield a triangular fuzzy number), and (3) progressive increase of fuzziness with successive operations. These limitations restrict the practical applicability of fuzzy numbers.

In this paper, we address these issues by introducing arithmetic based on extensional fuzzy numbers. We define fundamental arithmetic operations and relational operators ($=$, $>$, $\geq$, $<$, $\leq$) for this representation. The proposed approach provides closed-form operations without the need for discretisation or iterative numerical procedures, preserves the structure of fuzzy numbers under repeated computations, and prevents uncontrolled growth of fuzziness.

As a result, extensional fuzzy numbers can be directly used in mathematical models and standard algorithms as substitutes for real-valued quantities, without requiring algorithmic modifications. The usefulness of the approach is demonstrated through several application examples. A C++ implementation is available in a public GitHub repository.

1. Introduction

Fuzzy numbers are commonly represented with fuzzy sets. Their objective is to better handle imprecision of data. There are multiple representations of fuzzy numbers, eg. interval, triangular fuzzy sets. Linguistic interpretability of fuzzy numbers is their huge advantage. They can be used to represent linguistic terms (eg. "more or less 7", "almost 4", "slightly more than 5"). Description of numbers with linguistic terms is very common in humans. In everyday practice, we use linguistic numbers more frequently than precise numbers. One more fact is very important. Humans not only understand linguistic terms for numbers, but can also operate on them. We can, without any effort, sum up imprecise linguistic numbers (eg. circa 5 and more or less 4).

Linguistic numbers can be easily represented with fuzzy numbers. But operations on fuzzy numbers are not as straightforward as maths on crisp numbers. Commonly, Zadeh's extension rule is applied to elaborate a result. However, the application of this rule can produce three problems: (1) high computational expense: for some fuzzy sets, iterative numerical procedures are needed; (2) for some fuzzy sets and some operations, the result is not a fuzzy set with the same features (eg. multiplication of two triangular fuzzy sets does not produce a triangular fuzzy set);

(3) multiple arithmetical operation on fuzzy numbers may result in increase in fuzziness of fuzzy numbers (spread of fuzziness) beyond reasonable interpretability.

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These facts can severely limit the application field of fuzzy numbers. Fuzzy number find multiple applications. We highlight here two of them.

Commonly, fuzzy systems operate on type-1 fuzzy sets. However, not all types of imprecision of data can be represented with type-1 fuzzy sets, eg. intrapersonal and interpersonal uncertainties [1] cannot be handled with type-1 fuzzy sets [2]. This is why fuzzy systems with type-2 fuzzy sets were proposed [3]. In this case, the firing strengths of premises of fuzzy rules are type-1 fuzzy sets. The application of fuzzy sets (fuzzy numbers) in consequences requires arithmetic on fuzzy numbers [4]. This can be simplified by application interval type-2 fuzzy sets (IT2FSs) [1]. IT2FS-based systems have firing strengths of premises in the form on intervals [5–8]. Although, operations on intervals are simpler than on general fuzzy sets, they still manifest some problems. The main one is the spread of fuzziness in results. This is why in practice, the application of IT2FSs is limited and they are applied in premises and consequences operate on crisp values [8] or consequences are interval constants [5,9]. The type reduction is not trivial and commonly requires iterative numerical methods (eg. Karnik-Mendel procedure [10]). More reducers have been proposed [11–14] but they are not always consistent with the extension principle.

One more example of application of fuzzy numbers are graph algorithms with fuzzy weights [15–19]. Such graphs are used to represent imprecise data. In [16], they model the tourist flow. For navigation tasks, the shortest paths problem is sometimes discussed in terms of the time needed to complete the path rather than its length. The travel time is a variable with significant variance and this is why it is represented with fuzzy numbers [20]. Commonly, the shortest path problem is solved on graphs with triangular or trapezoidal fuzzy numbers as weights with α -cuts [19]. This limitation is caused by the problems with operations on fuzzy numbers.

In this paper, we would like to address the arithmetic of fuzzy numbers with a different kind of fuzzy numbers – extensional fuzzy numbers. We discuss the construction and properties of extensional fuzzy numbers. Then, we focus on relational operators ($=$, $>$, $\geq$, $<$, $\leq$) for extensional fuzzy numbers. Finally, we provide some illustrative examples of applications.

2. Related research

2.1. Granular computing

Lofti Zadeh coined a new paradigm for granular computing in his seminal paper in 1979 [21]. After a very long hibernation, this approach came back to life and made huge progress [22]. Today, when we have many techniques, methods, paradigms, algorithms, and experience [23], granular computing (GrC) is claimed a new emerging field of research in data mining and machine learning [24].

The main idea of granular computing is to provide ideas, techniques, and methods to make computers work in a way similar to human cognition [25]. This comprises granulation (decomposition of a whole into parts with preserved semantics), organisation (of parts into a whole), and causation (cause-effect relation). The idea of Zadeh was to make computer calculate with words [26,27]. GrC is also a starting point for a human-centric data approach [28].

To make the application of GrC easier, Yao [29] proposed a three-way approach to granular computing. His triarchic theory of GrC is based on three perspectives, each supporting the other two. First, the philosophical perspective handles structured thinking with the meronym-holonym concept (part-whole relation). Second, the methodological perspective provides structured problem solving with methods, algorithms, and techniques. Third, the computational perspective focuses on information processing.

Granular computing needs granules. Thus, the first step is the granulation of the provided data. To keep the semantics of elaborated granules, the principle of justifiable granulation is applied [30]. We can handle this step quite well. The next step is computing with granules. This is still a huge challenge for researchers. In this paper, we would like to propose an idea and technique for efficient operations of fuzzy granules.

2.1.1. Information granules

Information granule is a collection of entities in the sense of similarity, likeness, proximity, indiscernibility, identity, or adjacency [31–37]. An important feature of granules is their semantics. They can be labelled with semantically rich labels. This makes them a good representation of linguistic terms that are commonly used by humans. The second feature of granules is also human-centric: it is a hierarchy of granules. Granules can be split into subgranules when needed. Granules can merge into a higher level supergranules [38–40].

2.1.2. Representation of granules

Common representations are intervals, sets, fuzzy sets, interval type-2 fuzzy sets [41], rough sets [42,43], intuitionistic sets, shadowed sets, clusters [44], soft sets [45], and if-then rules [46]. A set is a simple representation of a collection of related items. Fuzzy sets enable partial membership, whereas rough sets address a case when no “yes-no decision” can be made. They are a starting point for the three-way decision approach [47–49].

The representation of granules is heavily combined with the granulation process. For example, granules represented with fuzzy sets may be elaborated with fuzzy clustering. Granules in the form of intervals are commonly produced with discretisation of the task domain. Typically, the techniques applied to elaborate granules are discretization, aggregation, clustering [50], and transformation [51].

2.2. Fuzzy numbers

Granular computing (GrC) aims at calculation with granules. This is why fuzzy numbers are used in GrC. If a fuzzy granule can be treated as a fuzzy number and we know how to operate on fuzzy numbers, we can calculate with fuzzy granules.

Below, we present a common definition of fuzzy numbers.

Definition 2.1 ([52]). A fuzzy number is a fuzzy set A that satisfies the following:

- Its membership function u_A is upper semicontinuous;
- $u_A(x) = 0$ outside some interval $[c, d]$;
- There are real numbers $a, b : c \leq a \leq b \leq d$ for which
 - u_A is monotonic increasing on $[c, a]$;
 - u_A is monotonic decreasing on $[b, d]$;
 - $u_A(x) = 1$ for $a \leq x \leq b$.

There are more definitions of fuzzy numbers [53]. These definitions are very general and allow for a very various construction of fuzzy numbers.

2.2.1. Representation of fuzzy numbers

In this section, we focus on operations on fuzzy numbers. They are defined with Zadeh's extension principle. The operation $*$ on fuzzy numbers results in [54]:

$$C(z) = A(x) \otimes B(y) = \sup_{z=x*y} \min(A(x), B(y)). \quad (1)$$

The realisation of Eq. (1) depends on the representation of fuzzy numbers. There are multiple representations proposed and used.

Intervals. An interval is a very simple example of a fuzzy number.

An interval fuzzy number is just an interval $[l, r]$. Operation $\otimes$ on an interval is defined as [55]:

$$[l_1, r_1] \otimes [l_2, r_2] = \{g_1 \otimes g_2 : l_1 \leq g_1 \leq r_1, l_2 \leq g_2 \leq r_2\}. \quad (2)$$

This definition is based on Zadeh's extension principle. In general, this definition requires discretisation and iterative elaboration of approximate results. However, for addition, subtraction, multiplication, and division closed-form formulae can be proved.

Triangular fuzzy numbers.

Definition 2.2. A triangular fuzzy number (l, c, r) is defined as a fuzzy set defined with three values $l \leq c \leq r$, where (l, r) is the support of the set and c is its core.

The general approach to operations on fuzzy numbers can be presented in the following steps:

- discretisation into a finite number of α -cuts,
- operations on α -cuts,
- merging of α -cuts into a final result.

Definition 2.3. The α -cut of a fuzzy set A is defined as a crisp set of all elements whose membership to the A set is not less than α :

$$A_\alpha = \{x \in X : u_A(x) \geq \alpha\}. \quad (3)$$

A sum of two triangular fuzzy numbers $A = (l_a, c_a, r_a)$ and $B = (l_b, c_b, r_b)$ is a triangular fuzzy number $A + B = (l_a + l_b, c_a + c_b, r_a + r_b)$ [56].

Scalar multiplication of a triangular fuzzy number $A = (l_a, c_a, r_a)$ for a crisp multiplier $k \geq 0$ is a triangular fuzzy number $kA = (kl_a, kc_a, kr_a)$ [56].

However, the set of triangular fuzzy sets is not closed under multiplication and division. The paper [57] presents a detailed analysis of operations on triangular fuzzy numbers. The authors prove that a product and a quotient of triangular fuzzy numbers are not triangular fuzzy numbers. Thus, triangular fuzzy numbers are not closed under multiplication and division. This is a problem when we want to execute multiple operations on triangular fuzzy numbers (eg. in a loop). Fig. 1(a) presents an example of multiplication of a triangular fuzzy number $x = (0, 1, 2)$ with itself (x^2 , x^3 , and x^3). The result is elaborated with 2000 α -cuts. The results are not triangular fuzzy numbers.

Gaussian fuzzy numbers. Application of Zadeh's extension principle to operations on Gaussian fuzzy sets is a decisively more complicated task than for triangular fuzzy sets. There are very few articles on this issue. The paper [58] presents some arithmetic on Gaussian fuzzy numbers but without any mathematical background. Fig. 1(b) presents an example of multiplication of a Gaussian fuzzy number $x = (1, 1)$ with itself. The result is elaborated with 2000 α -cuts. The result is very fuzzy, which makes it hardly applicable in practice.

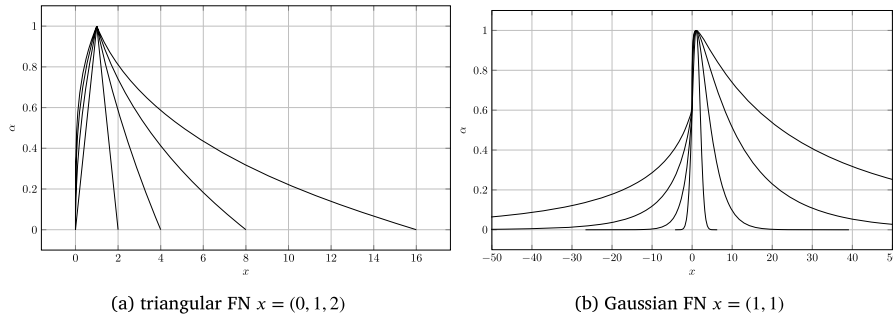


Fig. 1. Example of multiplication of fuzzy numbers (FN) with Zadeh's extension principle and 2000 α -cuts. The plots present the number x and its products: x^2 , x^3 , and x^4 .

2.3. Disadvantages of classical fuzzy numbers

Application of Zadeh's extension principle causes three cardinal problems:

- computational expense (in general, a discretisation and subsequent iterative approach are needed),
- the results do not preserve the shape of operands (it makes elaboration of results in loops difficult or even impossible),
- accumulation of fuzziness (the more operations are executed, the more fuzzy the result) [59].

These three problems make the application of fuzzy numbers based on Zadeh's extension principle hardly acceptable in practice. All these problems can be observed in Fig. 1. This is why in this paper, we would like to focus on a different approach to fuzzy numbers.

2.4. Summary

Fuzzy numbers are used to represent imprecise or linguistic numerical information and are commonly modelled using fuzzy sets such as intervals or triangular fuzzy numbers. Although they allow computers to process data in a way closer to human reasoning, performing arithmetic operations on them is more complex than with precise numbers. Commonly, Zadeh's extension principle is used to compute results but this approach often leads to iterative procedures and may produce results that do not preserve the original form of the fuzzy numbers (e.g., multiplying two Gaussian fuzzy numbers does not yield a Gaussian fuzzy number). Additionally, repeated operations can increase the fuzziness of results, limiting practical applications.

3. Extensional fuzzy numbers

This approach is based on a different approach from Zadeh's extension principle. The main idea here is an extension of a crisp number with a T-norm-similarity relation. Here, we only provide the essentials for understanding extensional fuzzy numbers. For details and mathematical background, please refer to [60–63].

Definition 3.1. A fuzzy relation R between X and Y is a fuzzy set

$$R(X, Y) = \{u(x, y)/(x, y) : x \in X \wedge y \in Y\}, \quad (4)$$

where u stands for the membership function.

Fuzzy $\star$ -similarity relation is a crucial notion for extensional fuzzy numbers and is defined as follows:

Definition 3.2 ([60]). A fuzzy $\star$ -similarity relation defined for a left continuous t-norm $\star$ satisfies the following properties for all $x, y, z \in \mathbb{R}$:

- reflexivity:

$$S(x, x) = 1, \quad (5)$$

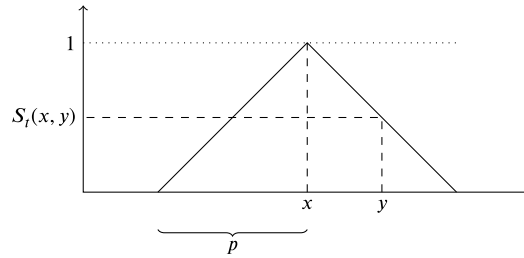
- symmetry:

$$S(x, y) = S(y, x), \quad (6)$$

- triangle property:

$$S(x, y) \star S(y, z) \leq S(x, z). \quad (7)$$

It is used to construct extensional fuzzy numbers.

Fig. 2. Fuzzy $\star$ -similarity relation S_t .

Definition 3.3 ([60]). The extensional fuzzy numbers are constructed with extension of crisp numbers with a fuzzy $\star$ -similarity relation defined for a left continuous t-norm $\star$. It can be proved that extensional fuzzy number x_S of x with regard to $\star$ -similarity relation S on $\mathbb{R}$ can be expressed as [60]:

$$\forall_{y \in \mathbb{R}} \quad x_S(y) = S(x, y). \quad (8)$$

Extensional fuzzy numbers have several advantages [60]:

- Multiple operations of fuzzy numbers do not increase the vagueness of fuzzy numbers.
- The result of an operation on extensional fuzzy numbers is a extensional fuzzy number.
- It is possible to use fuzzy numbers with an unbounded support (eg. for the Gaussian fuzzy sets).
- The computational overload of extensional fuzzy numbers is very low: no discretisation, no integration, no iterations are needed.
- A plethora of multiple types of extensional fuzzy numbers can be defined with fuzzy $\star$ -similarity relations.

3.1. Examples

Example 3.4. Let's define a fuzzy $\star$ -similarity relation S_t

$$S_t(x, y) = \max \left\{ 1 - \frac{|x - y|}{p}, 0 \right\}, \quad (9)$$

where $p > 0$ with the Łukasiewicz t-norm $\star$

$$T(x, y) = \max\{0, x + y - 1\}. \quad (10)$$

Fig. 2 presents a graphical representation of the extensional fuzzy number constructed with S_t .

Theorem 1. The relation S_t (Eq. (9)) is a fuzzy $\star$ -similarity relation for the Łukasiewicz t-norm (Eq. (10)).

Proof. The properties: reflexivity (Eq. (5)) and symmetry (6) are a direct conclusion from the definition of the S_t relation. The triangle property (Eq. (7)) can be proved with simple mathematics. $\square$

Example 3.5. Let's define a fuzzy $\star$ -similarity relation S_r

$$S_r(x, y) = \min \left\{ 1, \max \left\{ 1 - \frac{|x - y|}{p}, 0 \right\} \cdot \frac{1}{1 - k} \right\}, \quad (11)$$

where $p > 0$ and $0 \leq k < 1$ with the Łukasiewicz t-norm $\star$

$$T(x, y) = \max\{0, x + y - 1\}. \quad (12)$$

Fig. 3 presents a graphical representation of the extensional fuzzy number constructed with S_r .

Example 3.6. Let's define a fuzzy $\star$ -similarity relation S_e as:

$$S_e(x, y) = \exp \left(-\frac{|x - y|}{p} \right) \quad (13)$$

where $p > 1$ for the product t-norm $\star$. Fig. 4 presents a graphical representation of the extensional fuzzy number constructed with S_e .

Theorem 2. The relation S_e (Eq. (13)) is a fuzzy $\star$ -similarity relation for the product t-norm $\star$.

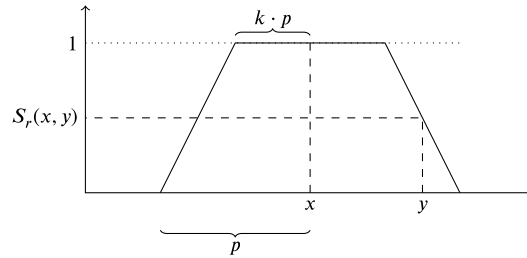
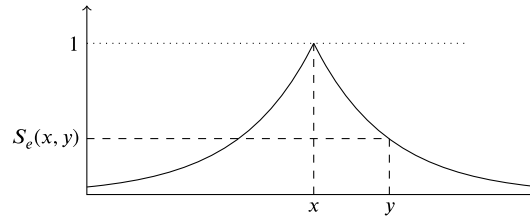
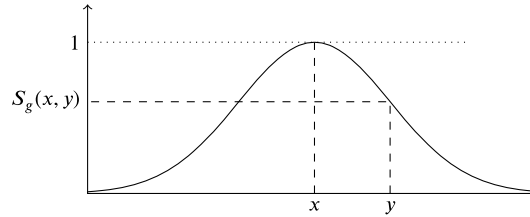
Proof. The properties: reflexivity (Eq. (5)) and symmetry (6) are a direct conclusion from the definition of the S_e relation. The triangle property (Eq. (7)) can be proved as below:

$$S_e(x, y) \star S_e(y, z) \leq S_e(x, z) \quad (14)$$

$$e^{-\frac{|x-y|}{p}} \times e^{-\frac{|y-z|}{p}} \leq e^{-\frac{|x-z|}{p}} \quad (15)$$

$$|x - y| + |y - z| \geq |x - z| \quad (16)$$

$\square$

Fig. 3. Fuzzy $\star$ -similarity relation S_r .Fig. 4. Fuzzy $\star$ -similarity relation S_e for $p = 1$.Fig. 5. Fuzzy $\star$ -similarity relation S_g for $p = 1$.

Example 3.7. Let's define a fuzzy $\star$ -similarity relation S_g as:

$$S_g(x, y) = \exp\left(-\frac{(x-y)^2}{2p^2}\right) \quad (17)$$

where $p > 0$ for the product t-norm. Fig. 5 presents a graphical representation of the extensional fuzzy number constructed with S_g .

Theorem 3. The relation S_g (Eq. (17)) is a fuzzy $\star$ -similarity relation for the product t-norm.

Proof. The properties: reflexivity (Eq. (5)) and symmetry (6) are a direct conclusion from the definition of the S_g relation. The triangle property (Eq. (7)) can be proved with simple mathematics.

$$S_g(x, y) \star S_g(y, z) \leq S_g(x, z) \quad (18)$$

$$\exp\left(-\frac{(x-y)^2}{2p^2}\right) \times \exp\left(-\frac{(y-z)^2}{2p^2}\right) \leq \exp\left(-\frac{(x-z)^2}{2p^2}\right) \quad (19)$$

$$(x-y)^2 + (y-z)^2 \geq (x-z)^2 \quad (20)$$

□

Remark 3.8. Extensional fuzzy numbers constructed with relation S_g and S_e do not satisfy the classical definition of fuzzy numbers because their supports are infinite. "However, we do not see any theoretical nor conceptual reason for that exclusion and we are convinced that this restriction is only of a technical nature and stems from the α -cut based arithmetic." [60].

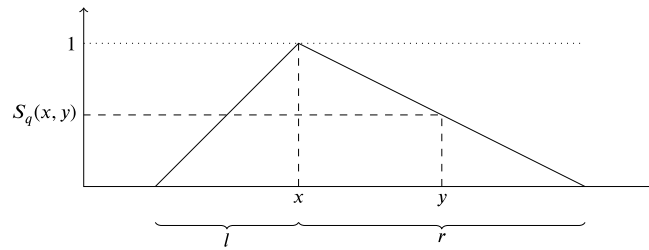
For the application example (Section 4.2), we also define a fuzzy $\star$ -quasi-similarity relation.

Definition 3.9. A fuzzy $\star$ -quasi-similarity relation defined for a left continuous t-norm $\star$ satisfies the following properties for all $x, y, z \in \mathbb{R}$:

$$S(x, x) = 1, \quad (21)$$

$$S(x, y) \star S(y, z) \leq S(x, z). \quad (22)$$

The difference between a fuzzy $\star$ -quasi-similarity and a fuzzy $\star$ -similarity is that the former is not symmetric.

Fig. 6. Fuzzy $\star$ -quasi-similarity relation S_q .

Example 3.10. Let's define a fuzzy $\star$ -quasi-similarity relation S_q

$$S_q(x, y) = \begin{cases} \max \left\{ 1 - \frac{|x-y|}{l}, 0 \right\}, & x > y \\ \max \left\{ 1 - \frac{|x-y|}{r}, 0 \right\}, & x \leq y \end{cases} \quad (23)$$

where $l > 0$ and $r > 0$ with the Łukasiewicz t-norm $\star$. Fig. 6 presents a graphical representation of the extensional fuzzy number constructed with S_q .

3.2. Arithmetic with extensional fuzzy numbers

The paper [60] proves arithmetical operations on extensional fuzzy numbers. For $S \in \{S_l, S_r, S_e, S_g\}$:

$$x_{S(p_x)} + y_{S(p_y)} = (x + y)_{S(\max\{p_x, p_y\})}, \quad (24)$$

$$x_{S(p_x)} \cdot y_{S(p_y)} = (x \cdot y)_{S(\max\{p_x, p_y\})}, \quad (25)$$

$$-(x_S) = (-x)_S, \quad (26)$$

$$(x_S)^{-1} = (1/x)_S, \text{ for } x_S \neq 0_S. \quad (27)$$

In (24) and (25), the max operator is used. The relations S_l , S_r , S_e , and S_g are defined in such a way that max can be used. In general, a closure of two relations is used [60].

A very interesting feature of extensional fuzzy numbers needs highlighting. In extensional fuzzy numbers, there is no unique zero and one. There are multiple zeroes and ones. All zeroes are constructed as an extension of $x = 0$ with different vagueness. The same applies to one. The set of extensional fuzzy numbers with addition, multiplication, identity elements 1 and 0 is not an algebraic field, it is a MI-monoid (many identities monoid) [60]. It may seem strange, but it is completely intuitive, because we humans understand a fuzzy zero, eg. "more or less zero", "precisely zero". It is also counterintuitive that arithmetic manipulation with vague values should have crisp 0 and crisp 1. For details and mathematical background, please refer to [60].

We use the same approach as [60] to define addition and multiplication of extensional fuzzy numbers constructed with a fuzzy $\star$ -quasi-similarity relation. Thus:

$$x_{S_q(l_x, r_x)} + y_{S_q(l_y, r_y)} = (x + y)_{S_q(\max\{l_x, l_y\}, \max\{r_x, r_y\})}, \quad (28)$$

$$x_{S_q(l_x, r_x)} \cdot y_{S_q(l_y, r_y)} = (x \cdot y)_{S_q(\max\{l_x, l_y\}, \max\{r_x, r_y\})}. \quad (29)$$

3.3. Comparison operators for extensional fuzzy numbers

The paper [63] defines the $\leq_S$ operator for extensional fuzzy numbers to return either an extensional fuzzy number for zero 0_S or an extensional fuzzy number for one 1_S . In this paper, we need an operator that returns a value from an interval $[0, 1]$. In the following section, we define $=_S, <_S, \leq_S, >_S, \geq_S$ operators for extensional fuzzy numbers.

Definition 3.11. A fuzzy binary relation R is any fuzzy set of a Cartesian product of two sets A, B of fuzzy objects:

$$R(A, B) \subseteq \{u(a, b)/(a, b) : a \in A \wedge b \in B\}. \quad (30)$$

Definition 3.12. The equality operator $=_S: \mathbb{E} \times \mathbb{E} \rightarrow [0, 1]$, where $\mathbb{E}$ is a set of extensional fuzzy numbers, is defined with the fuzzy $\star$ -similarity relation S is a value of the S relation for the bases of the extensional fuzzy numbers:

$$(x_{S(p_x)} =_S y_{S(p_y)}) = S_{(\max\{p_x, p_y\})}(x, y). \quad (31)$$

Equality relation is a fuzzy relation and its value is a number from an interval $[0, 1]$. Definition 3.12 makes the equality ($=_S$) transitive, because the fuzzy $\star$ -similarity relation S satisfies the condition (7).

We use the $=_S$ relation for extensional fuzzy numbers to define $<_S, >_S, \leq_S, \geq_S$ relations.

Definition 3.13. The operator $<_S$ for two extensional fuzzy numbers is defined as

$$x_{S(p_x)} <_S y_{S(p_y)} = \begin{cases} 1 - (x_{S(p_x)} =_S y_{S(p_y)}), & x \leq y, \\ 0, & x > y. \end{cases} \quad (32)$$

Definition 3.14. The operator $>_S$ for two extensional fuzzy numbers is defined as

$$x_{S(p_x)} >_S y_{S(p_y)} = \begin{cases} 1 - (x_{S(p_x)} =_S y_{S(p_y)}), & x \geq y, \\ 0, & x < y. \end{cases} \quad (33)$$

Definition 3.15. The operator $\leq_S$ for two extensional fuzzy numbers is defined as

$$x_{S(p_x)} \leq_S y_{S(p_y)} = \begin{cases} 1, & x \leq y, \\ (x_{S(p_x)} =_S y_{S(p_y)}), & x > y. \end{cases} \quad (34)$$

Definition 3.16. The operator $\geq_S$ for two extensional fuzzy numbers is defined as

$$x_{S(p_x)} \geq_S y_{S(p_y)} = \begin{cases} 1, & x \geq y, \\ (x_{S(p_x)} =_S y_{S(p_y)}), & x < y. \end{cases} \quad (35)$$

The definitions of comparison operators based on extensional fuzzy numbers are computationally efficient unlike the ordering operators proposed in [20] that require iterative procedures.

Many techniques for comparison of fuzzy numbers are based on iterative approach [64–66]. The examples below show the comparison of extensional fuzzy numbers. It is run straightforward without iterative procedures. The operators defined above return values from the interval $[0, 1]$ instead of binary 0 or 1.

Example 3.17. Let's define three extensional fuzzy numbers with the S_t and the Łukasiewicz t-norm for crisp values: $a = 0$ (for $p_a = 1$), $b = 2$ (for $p_b = 3$), $c = 3$ (for $p_c = 0.5$). The graphical representation of the numbers is presented in Fig. 7.

1. The value of $a_{S_t(p_a)} =_S b_{S_t(p_b)}$ is $S_{t(\max\{p_a, p_b\})}(a, b) = \max \left\{ 1 - \frac{|a-b|}{\max\{p_a, p_b\}}, 0 \right\} = \frac{1}{3}$. Respectively:

- $a_{S_t(p_a)} <_S b_{S_t(p_b)} = \frac{2}{3}$,
- $a_{S_t(p_a)} \leq_S b_{S_t(p_b)} = 1$,
- $a_{S_t(p_a)} >_S b_{S_t(p_b)} = 0$,
- $a_{S_t(p_a)} \geq_S b_{S_t(p_b)} = \frac{1}{3}$.

2. The value of $b_{S_t(p_b)} =_S c_{S_t(p_c)}$ is $S_{t(\max\{p_b, p_c\})}(b, c) = \max \left\{ 1 - \frac{|b-c|}{\max\{p_b, p_c\}}, 0 \right\} = \frac{2}{3}$. Respectively:

- $b_{S_t(p_b)} <_S c_{S_t(p_c)} = \frac{1}{3}$,
- $b_{S_t(p_b)} \leq_S c_{S_t(p_c)} = 1$,
- $b_{S_t(p_b)} >_S c_{S_t(p_c)} = 0$,
- $b_{S_t(p_b)} \geq_S c_{S_t(p_c)} = \frac{2}{3}$.

3. The value of $a_{S_t(p_a)} =_S c_{S_t(p_c)}$ is $S_{t(\max\{p_a, p_c\})}(a, c) = \max \left\{ 1 - \frac{|a-c|}{\max\{p_a, p_c\}}, 0 \right\} = 0$. Respectively:

- $a_{S_t(p_a)} <_S c_{S_t(p_c)} = 1$,
- $a_{S_t(p_a)} \leq_S c_{S_t(p_c)} = 1$,
- $a_{S_t(p_a)} >_S c_{S_t(p_c)} = 0$,
- $a_{S_t(p_a)} \geq_S c_{S_t(p_c)} = 0$.

Example 3.18. Let's define three extensional fuzzy numbers with the S_r and the Łukasiewicz t-norm for crisp values: $a = 0$ (for $p_a = 3$), $b = 3$ (for $p_b = 5$), $c = 7$ (for $p_c = 0.5$), $k = 0.5$. The numbers are presented in Fig. 8

1. The value of $a_{S_r(p_a)} =_S b_{S_r(p_b)}$ is by definition $S_{r(\max\{p_a, p_b\})}(a, b) = \min \left\{ 1, \max \left\{ 1 - \frac{|a-b|}{\max\{p_a, p_b\}}, 0 \right\} \cdot \frac{1}{1-k} \right\} = \frac{4}{5}$. Respectively:

- $a_{S_r(p_a)} <_S b_{S_r(p_b)} = \frac{1}{5}$,
- $a_{S_r(p_a)} \leq_S b_{S_r(p_b)} = 1$,
- $a_{S_r(p_a)} >_S b_{S_r(p_b)} = 0$,
- $a_{S_r(p_a)} \geq_S b_{S_r(p_b)} = \frac{4}{5}$.

2. The value of $b_{S_r(p_b)} =_S c_{S_r(p_c)}$ is by definition $S_{r(\max\{p_b, p_c\})}(b, c) = \min \left\{ 1, \max \left\{ 1 - \frac{|b-c|}{\max\{p_b, p_c\}}, 0 \right\} \cdot \frac{1}{1-k} \right\} = \frac{2}{5}$. Respectively:

- $b_{S_r(p_b)} <_S c_{S_r(p_c)} = \frac{3}{5}$,
- $b_{S_r(p_b)} \leq_S c_{S_r(p_c)} = 1$,

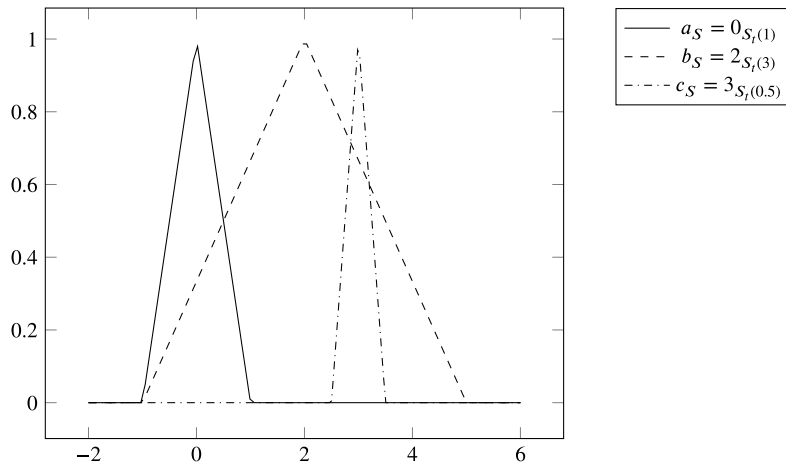


Fig. 7. Three extensional fuzzy numbers defined with the S_i $\star$ -similarity relation and the Łukasiewicz t-norm.

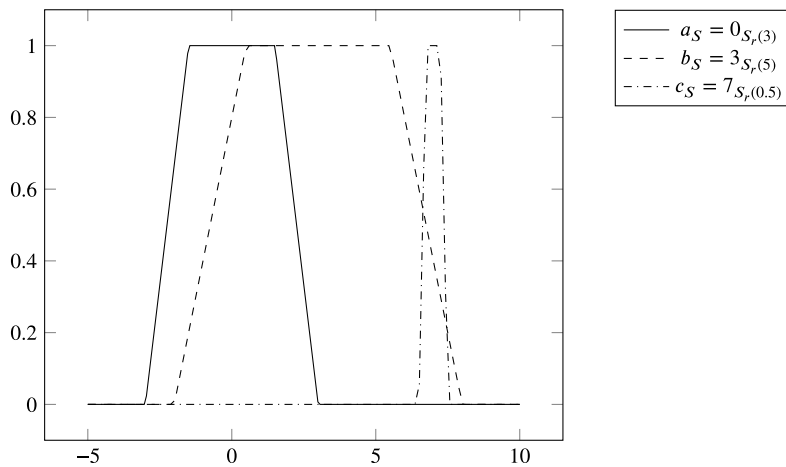


Fig. 8. Three extensional fuzzy numbers defined with the S_r fuzzy $\star$ -similarity relation and the Łukasiewicz t-norm.

- $b_{S_i(p_b)} >_S c_{S_i(p_c)} = 0$,
- $b_{S_i(p_b)} \geq_S c_{S_i(p_c)} = \frac{2}{5}$.

3. The value of $a_{S_i(p_a)} =_S c_{S_i(p_b)}$ is by definition $S_{r(\max\{p_a, p_b\})}(a, c) = \min \left\{ 1, \max \left\{ 1 - \frac{|a-c|}{\max\{p_a, p_b\}}, 0 \right\} \cdot \frac{1}{1-k} \right\} = 0$. Respectively:

- $a_{S_i(p_a)} <_S c_{S_i(p_c)} = 1$,
- $a_{S_i(p_a)} \leq_S c_{S_i(p_c)} = 1$,
- $a_{S_i(p_a)} >_S c_{S_i(p_c)} = 0$,
- $a_{S_i(p_a)} \geq_S c_{S_i(p_c)} = 0$.

Example 3.19. Let's define three extensional fuzzy numbers with the S_e and the product t-norm for crisp values: $a = 0$ (for $p_a = 1$), $b = 2$ (for $p_b = 3$), $c = 5$ (for $p_c = 0.5$). The numbers are presented in Fig. 9.

1. The value of $a_{S_e(p_a)} =_S b_{S_e(p_b)}$ is $S_{e(\max\{p_a, p_b\})}(a, b) = \exp \left(-\frac{|a-b|}{\max\{p_a, p_b\}} \right) \approx 0.513$. Respectively:

- $a_{S_e(p_a)} <_S b_{S_e(p_b)} \approx 1 - 0.513 = 0.487$,
- $a_{S_e(p_a)} \leq_S b_{S_e(p_b)} = 1$,
- $a_{S_e(p_a)} >_S b_{S_e(p_b)} = 0$,
- $a_{S_e(p_a)} \geq_S b_{S_e(p_b)} \approx 0.513$.

2. The value of $a_{S_e(p_a)} =_S c_{S_e(p_c)}$ is $S_{e(\max\{p_a, p_c\})}(a, c) = \exp \left(-\frac{|a-c|}{\max\{p_a, p_c\}} \right) \approx 0.0067$. Respectively:

- $a_{S_e(p_a)} <_S c_{S_e(p_c)} \approx 1 - 0.0067 = 0.9933$,
- $a_{S_e(p_a)} \leq_S c_{S_e(p_c)} = 1$,
- $a_{S_e(p_a)} >_S c_{S_e(p_c)} = 0$,
- $a_{S_e(p_a)} \geq_S c_{S_e(p_c)} \approx 0.0067$.

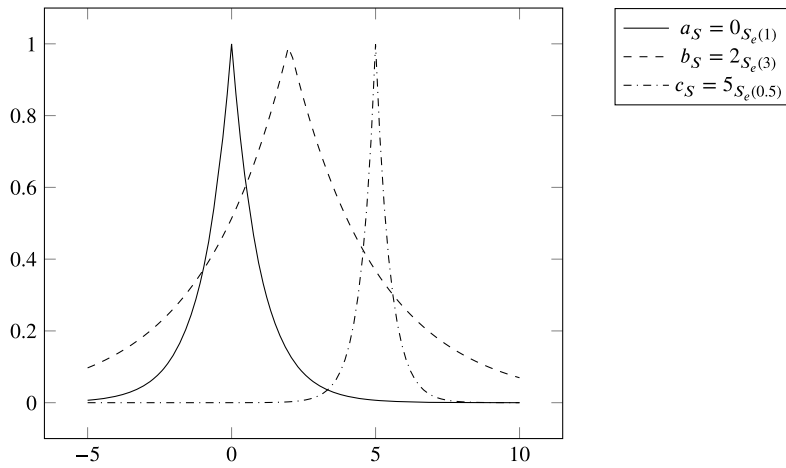


Fig. 9. Three extensional fuzzy numbers defined with the S_e fuzzy $\star$ -similarity relation and the Łukasiewicz t-norm.

3. The value of $b_{S_e(p_b)} =_S c_{S_e(p_c)}$ is $S_{e(\max\{p_b, p_c\})}(b, c) = \exp\left(-\frac{|b-c|}{\max\{p_b, p_c\}}\right) \approx 0.368$. Respectively:

- $b_{S_e(p_b)} <_S c_{S_e(p_c)} \approx 1 - 0.368 = 0.632$,
- $b_{S_e(p_b)} \leq_S c_{S_e(p_c)} = 1$,
- $b_{S_e(p_b)} >_S c_{S_e(p_c)} = 0$,
- $b_{S_e(p_b)} \geq_S c_{S_e(p_c)} \approx 0.368$.

Example 3.20. Let's define three extensional fuzzy numbers with the S_g and the product t-norm for crisp values: $a = 0$ (for $p_a = 1$), $b = 2$ (for $p_b = 2$), $c = 5$ (for $p_c = 0.5$). The numbers are presented in Fig. 10.

1. The value of $a_{S_g(p_a)} =_S b_{S_g(p_b)}$ is $S_{g(\max\{p_a, p_b\})}(a, b) = \exp\left(-\frac{(a-b)^2}{2(\max\{p_a, p_b\})^2}\right) \approx 0.8007$. Respectively:

- $a_{S_g(p_a)} <_S b_{S_g(p_b)} \approx 1 - 0.8007 = 0.1993$,
- $a_{S_g(p_a)} \leq_S b_{S_g(p_b)} = 1$,
- $a_{S_g(p_a)} >_S b_{S_g(p_b)} = 0$,
- $a_{S_g(p_a)} \geq_S b_{S_g(p_b)} \approx 0.8007$,

2. The value of $a_{S_g(p_a)} =_S c_{S_g(p_c)}$ is $S_{g(\max\{p_a, p_c\})}(a, c) = \exp\left(-\frac{(a-c)^2}{2(\max\{p_a, p_c\})^2}\right) \approx 0.000003727$. Respectively:

- $a_{S_g(p_a)} <_S c_{S_g(p_c)} \approx 1 - 0.000003727 = 0.999996273$,
- $a_{S_g(p_a)} \leq_S c_{S_g(p_c)} = 1$,
- $a_{S_g(p_a)} >_S c_{S_g(p_c)} = 0$,
- $a_{S_g(p_a)} \geq_S c_{S_g(p_c)} \approx 0.000003727$.

3. The value of $b_{S_g(p_b)} =_S c_{S_g(p_c)}$ is $S_{g(\max\{p_b, p_c\})}(b, c) = \exp\left(-\frac{(b-c)^2}{2(\max\{p_b, p_c\})^2}\right) \approx 0.6065$. Respectively:

- $b_{S_g(p_b)} <_S c_{S_g(p_c)} \approx 1 - 0.6065 = 0.3935$,
- $b_{S_g(p_b)} \leq_S c_{S_g(p_c)} = 1$,
- $b_{S_g(p_b)} >_S c_{S_g(p_c)} = 0$,
- $b_{S_g(p_b)} \geq_S c_{S_g(p_c)} \approx 0.6065$.

Example 3.21. Fig. 11 presents the comparison of the impact of the fuzzification parameter p on the value of the $=_S$ operator. The cores of the extensional fuzzy numbers are constants 70 and 80. The higher the value of p (fuzzification), more equal the numbers are. This is fully consistent with the intuition that if two non-identical fuzzy numbers are very fuzzy, they can be treated as almost equal and can be used to present the same fuzzy term. The equality operator $=_S$ can return 1 for two fuzzy numbers with different cores only for the trapezoidal S_r relation of four presented in this paper.

3.4. Summary

Extensional fuzzy numbers are constructed by extending crisp numbers using fuzzy t -norm similarity relations instead of the classical Zadeh extension principle. In this approach, the membership of a value is determined by its similarity to a central crisp value, defined with a fuzzy similarity relation that satisfies properties such as reflexivity, symmetry, and a triangle condition. Different similarity relations (e.g., triangular, trapezoidal, exponential, Gaussian, etc.) can be used to create various types of extensional fuzzy numbers, allowing flexible modelling of fuzziness, including numbers with unbounded support. Arithmetic operations on extensional fuzzy numbers (such as addition, multiplication, negation, and inversion) have simple analytical forms, preserve the extensional

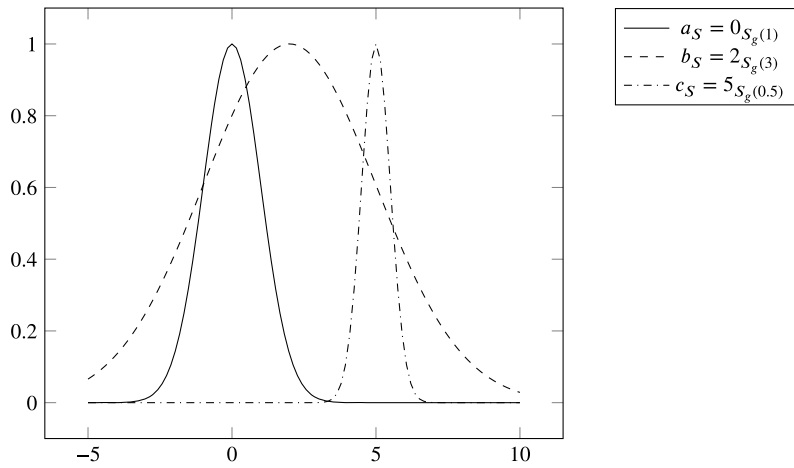


Fig. 10. Three extensional fuzzy numbers defined with the S_g fuzzy $\star$ -similarity relation and the Łukasiewicz t-norm.

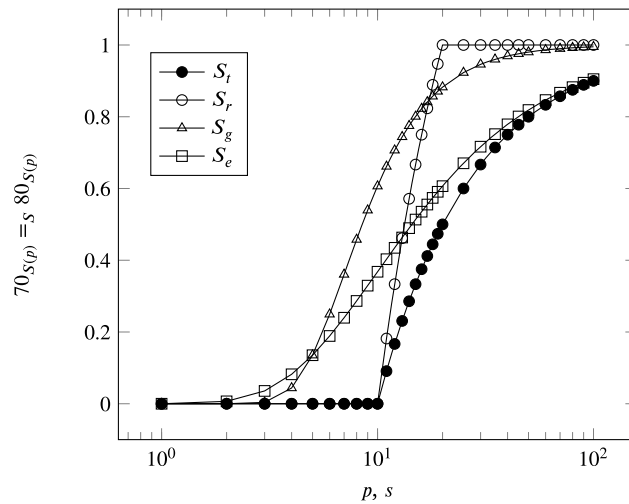


Fig. 11. Comparison of values of the relation $70_{S(p)} =_S 80_{S(p)}$ for various values of the p parameter and $S \in \{S_i, S_r, S_g, S_e\}$.

fuzzy number structure, do not suffer from the fuzzy spread, and do not require iterative procedures. The framework also introduces fuzzy comparison operators (equality and ordering relations) that return values from the interval $[0, 1]$, enabling graded comparisons between extensional fuzzy numbers based on their underlying similarity relations.

4. Application examples

The implementation of extensional fuzzy numbers and the source code for the examples below are available from the free public GitHub repository: github.com/ksiminski/ksi-extensional-fuzzy-numbers.

4.1. Sorting

Algorithm 1 presents a pseudocode for the insertion sort algorithm for extensional fuzzy numbers. The $<_S$ operator returns a value from the interval $[0, 1]$. To sort fuzzy numbers, a threshold ξ is needed because a decision has to be made whether to swap two extensional fuzzy numbers. The threshold is a parameter of the sorting algorithm.

The impact of the ξ threshold parameter can be seen in the following example.

Initial array of Gaussian extensional fuzzy numbers represented as pairs (core x and fuzziness p in Fig. 5):

$$(5.5, 0.5)(6, 3)(4.5, 1)(-4.3, 1)(5.6, 1)(0.4, 2)(7.7, 1)(4.2, 1.5).$$

The numbers are plotted in Fig. 12(a). The plot presents an array of Gaussian extensional fuzzy numbers represented as boxes with the core in the middle and the fuzziness as the upper and lower parts of the boxes.

Algorithm 1 Pseudocode of the insertion sort algorithm for extensional fuzzy numbers.

Require: $A[1..n]$: array of extensional fuzzy numbers
Require: ξ : threshold for the $<_S$ operator for extensional fuzzy numbers

```

1: procedure INSERTION_SORT( $A, \xi$ )
2:   for  $i \leftarrow 2$  to  $n$  do
3:      $minimum \leftarrow A[i]$ 
4:      $j \leftarrow i - 1$ 
5:     while  $j > 0$  and  $(minimum <_S A[j]) > \xi$  do
6:        $A[j + 1] \leftarrow A[j]$ 
7:        $j \leftarrow j - 1$ 
8:     end while
9:      $A[j + 1] \leftarrow minimum$ 
10:  end for
11: end procedure

```

The results of sorting depends on the values of ξ .

- $\xi = 0.0$: $(-4.3, 1)(0.4, 2)(4.2, 1.5)(4.5, 1)(5.5, 0.5)(5.6, 1)(6, 3)(7.7, 1)$
 For $\xi = 0$, the vagueness for fuzzy numbers does not influence the result of sorting. The values are sorted strictly according to their cores (Fig. 12(b)).
- $\xi = 0.5$: $(-4.3, 1)(0.4, 2)(5.5, 0.5)(6, 3)(4.5, 1)(5.6, 1)(4.2, 1.5)(7.7, 1)$
 In this case, some numbers (eg. $(6, 3)$ and $(4.5, 1)$) are not swapped in the sorting algorithm because of their significant vagueness. The values in array cells 3–8 are indistinguishable from each other in terms of the $<_S$ operator and the threshold ξ (Fig. 12(c)).
- $\xi = 0.9$: $(-4.3, 1)(5.5, 0.5)(6, 3)(4.5, 1)(0.4, 2)(5.6, 1)(4.2, 1.5)(7.7, 1)$
 For $\xi = 0.9$, even $(4.5, 1)$ in cell 4 and $(0.4, 2)$ in cell 5 of the array are not swapped in the sorting algorithm because of their significant fuzziness and high value of ξ . The values in array cells 2–8 are not distinguishable from each other (Fig. 12(d)) for this values of ξ .

4.2. The shortest paths in a fuzzy weighted graph

The problem stated in this section is a fuzzy version of the very well-known problem in weighted graphs. It is commonly solved with Dijkstra's algorithm (all paths from one node, no negative edges allowed), Bellman-Ford algorithm (all paths from one node, negative edges allowed), and Floyd-Warshall algorithm (all paths between any pair of vertices).

The problems in application of fuzzy numbers to the shortest path problem are [15]:

- costs without existing paths [15,20],
- a fuzzy set of nondominated paths [20] as a result,
- limitation to graphs with non-negative edge weights.

The main problem in this task is the ordering of fuzzy numbers. This is solved by assigning an index value to fuzzy numbers. This is not trivial because commonly indices need elaboration of integrals (Yager's index [64,65,67], Liou-Wang index [66], García-Lamata index [68]). Some techniques avoid integrals [20,69]. Dubois and Prade proposed the possibility index [70] and Nayeem and Pal the acceptability index [71]. Indices proposed by Yager, Liou and Wang, García and Lamata, Dubois, and Prade return a single solution. Algorithms based on the Okada and Soper index and the Nayeem and Pal acceptability index return a set of solutions. Some authors apply genetic algorithms [72,73], biologically based heuristic [17], artificial bee colony heuristic [19], ant colony heuristic [18], or dynamic programming [74,75].

The paper [76] presents a modification of Dijkstra's algorithm for the shortest path problem in graphs with interval type-2 fuzzy weights of edges. The algorithm ranks the centroids of interval type-2 fuzzy numbers to compare them. Type reduction is run with the iterative Karnik-Mendel algorithm [10].

In the following subsection (Section 4.2.1), we present our approach in comparison with the results elaborated by the techniques mentioned above [15,16]. We use quasi-similarity relation (Definition 3.9) to enable asymmetric triangular fuzzy numbers because all techniques above apply asymmetric triangular fuzzy numbers to represent edge weights in a graph. In the following examples, edge weights are asymmetric triangular fuzzy numbers in the form (m, α, β) , where m stands for the core, $(m - \alpha, m + \beta)$ is the support interval [15].

The cited paper presents two examples of the shortest path problem for a graph with triangular fuzzy weights of edges. Application of fuzzy numbers in [15] requires modification of the shortest path algorithm. The algorithm is based on ordering functions of fuzzy numbers. In the article, six functions are used (four of them require integrations). In the analysis of the computational complexity of the algorithm, the authors admit that the "Okada and Soper's order relation is the most expensive" time-unit operation [15].

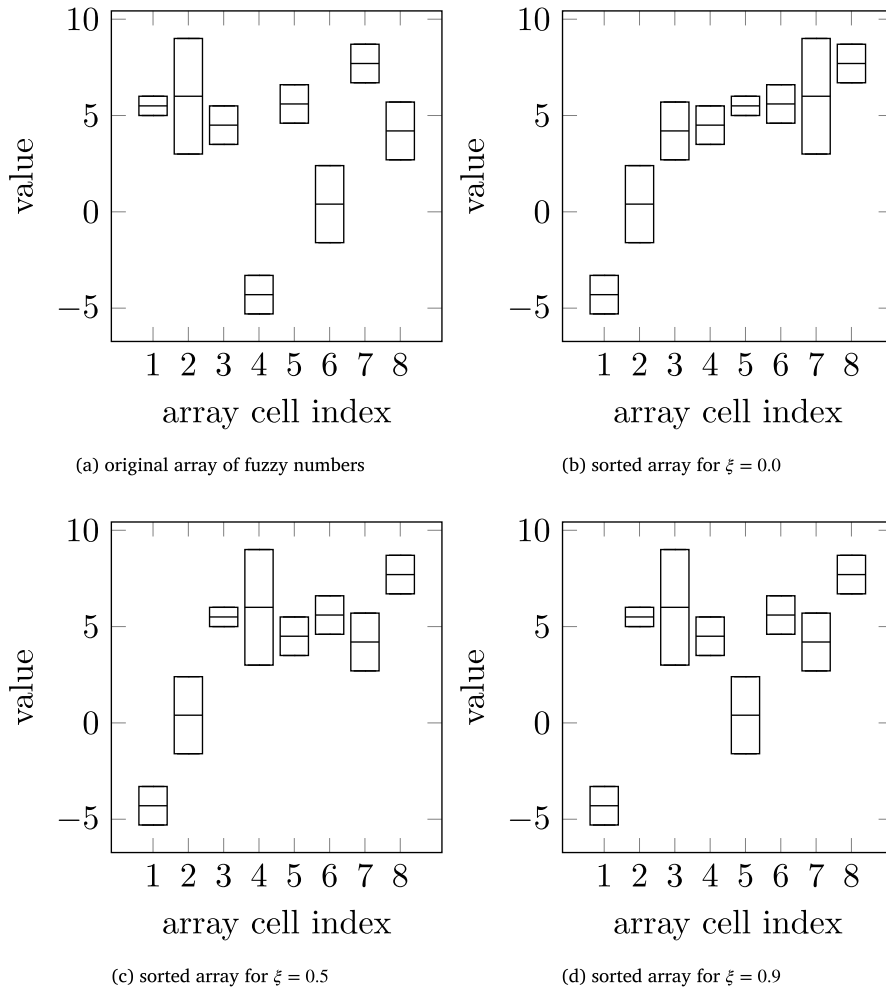


Fig. 12. Example of sorting of Gaussian extensional fuzzy numbers for various values of threshold ξ . The plots present arrays of Gaussian extensional fuzzy numbers represented by rectangles whose heights represent the fuzziness of numbers.

4.2.1. Comparison with other approaches

In this example, we compare the results of the Floyd-Warshall algorithm for the shortest path for extensional fuzzy numbers (Algorithm 2) with the results obtained in [15] for two graphs with fuzzy weight. The weights are asymmetrical triangular fuzzy sets. This is why we apply fuzzy $\star$ -quasi-similarity relation S_q (Definition 3.9). Extensional fuzzy numbers can be applied to the shortest path problem without modification of the algorithm and have minimal computational cost $O(1)$.

Two graphs taken from [15] are presented in Figs. 13 and 14. In both graphs node 1 is a starting node. The latter graph has an edge with a negative weight between nodes 4 and 5. The comparisons of results are presented in Tables 1 and 2, respectively. In both tables additional columns (EFN) presenting lengths of paths elaborated with extensional fuzzy numbers have been added.

For both graphs, our approach returns the shortest paths. In it worth mentioning that for some other methods, the paths are not always the shortest, eg. for the second graf (Fig. 14), the Okada-Soper approach elaborates longer paths than other techniques (Table 2).

4.2.2. Impact of fuzziness on the elaborated results

In examples presented in Section 4.2.1, our approach returns unique paths. This is because the fuzziness of edges is not fuzzy enough to influence the ordering of fuzzy numbers. In this example, we would like to show the influence of fuzziness on the elaborated paths. In the existing solutions, triangular [15–19] or trapezoidal [19] fuzzy sets for fuzzy edge weights are commonly used. In this example, we would like to show that our approach enables application of other fuzzy numbers as well. In this example, we use Gaussian

extensional fuzzy numbers. The graph is presented in Fig. 15. Fuzzy numbers are encoded as pairs of values: core x and fuzziness p (Fig. 5).

Algorithm 2 The Floyd-Warshall algorithm for the shortest paths between all pairs of nodes in a graph with fuzzy weights.

Require: $G = (\mathbb{V}, \mathbb{E})$: graph with the set $\mathbb{V}$ of vertices and the set $\mathbb{E}$ of edges

Require: w : edge weights

Require: ξ : threshold for the comparison operator

```

1:  $n \leftarrow |\mathbb{V}|$ ;
2: for  $i \leftarrow 1$  to  $n$  do
3:   for  $j \leftarrow 1$  to  $n$  do
4:      $p[i, j] \leftarrow j$ 
5:     if  $i = j$  then
6:        $d[i, j] \leftarrow 0$ 
7:     else if  $\exists w_{ij}$  then
8:        $d[i, j] \leftarrow w_{ij}$ 
9:     else
10:       $d[i, j] \leftarrow \infty$ 
11:       $p[i, j] \leftarrow \emptyset$ 
12:    end if
13:  end for
14: end for
15: for  $k \leftarrow 1$  to  $n$  do
16:   for  $i \leftarrow 1$  to  $n$  do
17:    if  $d[i, k] \neq \infty$  then
18:      for  $j \leftarrow 1$  to  $n$  do
19:        if  $(d[i, k] + d[k, j] <_S d[i, j]) > \xi$  then
20:           $d[i, j] \leftarrow d[i, k] + d[k, j]$ 
21:           $p[i, j] \leftarrow p[i, k]$ 
22:        end if
23:      end for
24:    end if
25:  end for
26: end for

```

▷ number of edges
▷ initialisation

▷ elaboration of distances

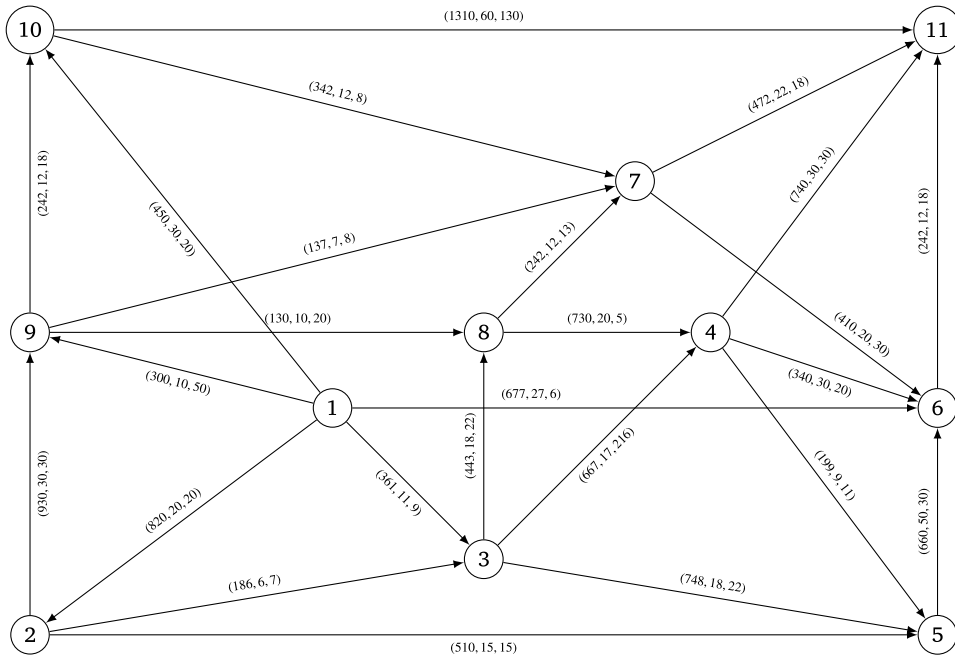


Fig. 13. Graph with fuzzy weights from Example 1 [15]. The shortest path for this graph are presented in Table 1.

Table 1

The comparison of the shortest paths elaborated for graph presented in Fig. 13. The order relations Yager (Y), Liou-Wang (LW), García-Lamata (GL), Okada-Soper (OS), Nayeem-Pal (NP), Dubois-Prade (DP) are taken from [15]. The column “EFN” (extensional fuzzy number) presents path length elaborated with our approach.

End node	Shortest path	Path cost		Order relation
		[15]	EFN	
2	1 → 2	(820, 20, 20)	(820, 20, 20)	all
3	1 → 3	(361, 11, 9)	(361, 11, 9)	all
4	1 → 3 → 4	(1028, 28, 225)	(1028, 17, 216)	all
4	1 → 9 → 8 → 4	(1167, 37, 63)	n/a	OS ($\epsilon = 0$)
5	1 → 3 → 5	(1109, 29, 31)	(1109, 18, 22)	all
6	1 → 6	(677, 27, 6)	(677, 27, 6)	all
7	1 → 9 → 7	(430, 20, 70)	(430, 10, 50)	all
8	1 → 9 → 8	(437, 17, 58)	(437, 10, 50)	all
9	1 → 9	(300, 10, 50)	(300, 10, 50)	all
10	1 → 10	(450, 30, 20)	(450, 30, 20)	all
11	1 → 9 → 7 → 11	(902, 42, 88)	(902, 22, 50)	Y, LW ($\lambda = 1$), GL ($\lambda = 1, \delta = 0$), OS ($\epsilon = 0, \epsilon = 0.5$), EFN
11	1 → 6 → 11	(919, 39, 24)	n/a	LW ($\lambda = 0, \lambda = 0.5$), GL ($\lambda \neq 1, \delta \neq 0$), NP, DP

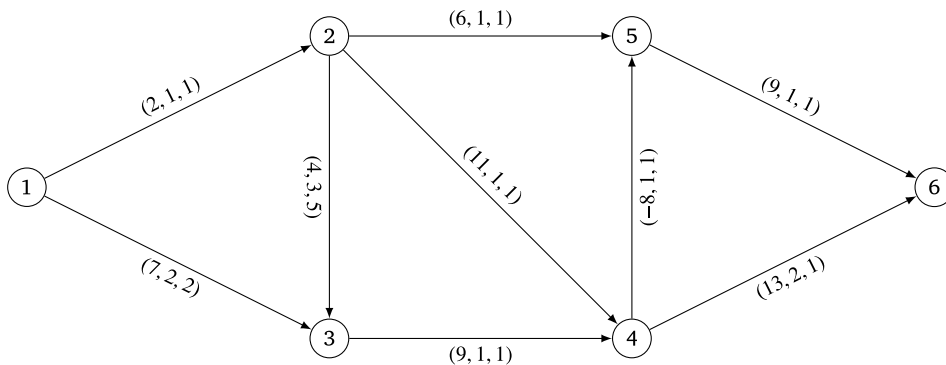


Fig. 14. Graph with fuzzy weights z [15]. The shortest path for this graph are presented in Table 2.

Table 2

The comparison of the shortest paths elaborated for graph presented in Fig. 14. The order relations Yager (Y), Liou-Wang (LW), García-Lamata (GL), Okada-Soper (OS), Nayeem-Pal (NP), Dubois-Prade (DP) are taken from [15]. The column “EFN” (extensional fuzzy number) presents path length elaborated with our approach.

End node	Shortest path	Path cost		Order relation
		[15]	EFN	
2	1 → 2	(2, 1, 1)	(2, 1, 1)	all
3	1 → 3	(7, 2, 2)	n/a	OS ($\epsilon = 0$ and 0.5), LW ($\lambda = 1$), GL ($\lambda = 1, \delta = 0.5$)
3	1 → 2 → 3	(6, 4, 6)	(6, 3, 5)	all, except LW ($\lambda = 1$), GL ($\lambda = 1, \delta = 0$)
4	1 → 2 → 4	(13, 2, 2)	(13, 1, 1)	all
4	1 → 2 → 3 → 4	(15, 5, 7)	n/a	OS ($\epsilon = 0$)
5	1 → 2 → 4 → 5	(5, 3, 3)	(5, 1, 1)	all
5	1 → 2 → 3 → 4 → 5	(7, 6, 8)	n/a	OS ($\epsilon = 0$)
6	1 → 2 → 4 → 5 → 6	(14, 4, 4)	(14, 1, 1)	all
6	1 → 2 → 3 → 4 → 5 → 6	(16, 7, 9)	n/a	OS ($\epsilon = 0$)

The fuzziness of fuzzy numbers has significant impact on the results. The thresholding enables various influences of fuzziness on the final result.

As an example, we would like to presents a few paths elaborated for various values of ξ .

- The path from node 4 to 6
 - for $\xi = 0.01$ with length (17, 8) is $4 \xrightarrow{(17,8)} 6$;
 - for $\xi = 0.001$ with length (16, 9) is $4 \xrightarrow{(5,9)} 1 \xrightarrow{(6,8)} 3 \xrightarrow{(5,7)} 6$;
- The path from node 7 to 6
 - for $\xi = 0.01$ with length (30, 8) is $7 \xrightarrow{(5,8)} 5 \xrightarrow{(8,8)} 4 \xrightarrow{(17,8)} 6$;
 - for $\xi = 0.001$ with length (28, 9) is $7 \xrightarrow{(5,8)} 5 \xrightarrow{(5,8)} 2 \xrightarrow{(2,9)} 4 \xrightarrow{(5,9)} 1 \xrightarrow{(6,8)} 3 \xrightarrow{(5,7)} 6$;

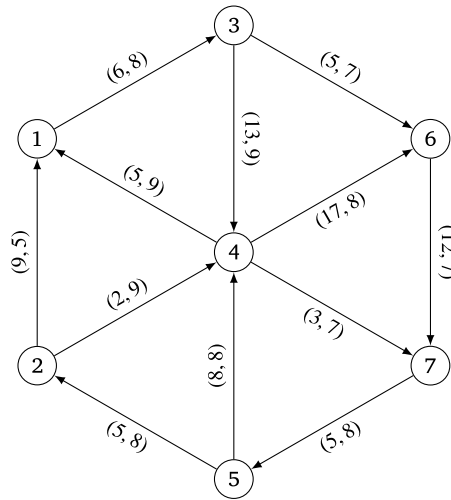


Fig. 15. Graph with fuzzy weights.

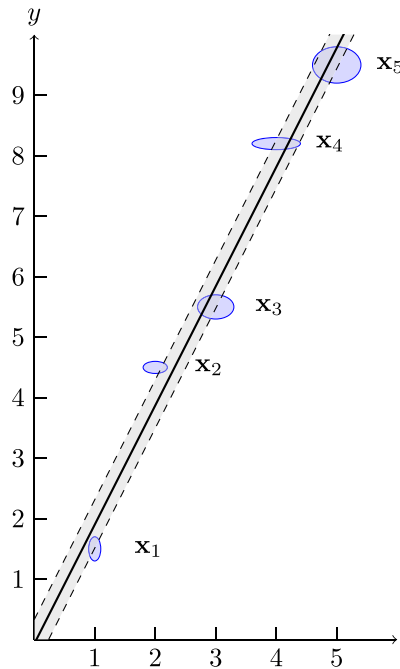


Fig. 16. Linear regression for two dimensional Gaussian extensional fuzzy numbers $x_1, \dots, x_5$. The ellipses represent standard deviations of fuzzy numbers. The regression line is built by the cores of the Gaussian extensional fuzzy numbers representing the fuzzy regression line. The line is accompanied by two dashed lines representing the standard deviation of Gaussian extensional fuzzy numbers of the regression line.

- The path from node 2 to 1
 - for $\xi = 0.1$ with length $(9, 5)$ is $2 \xrightarrow{(9,5)} 1$;
 - for $\xi = 0.01$ with length $(7, 9)$ is $2 \xrightarrow{(2,9)} 4 \xrightarrow{(5,9)} 1$;

It is worth highlighting that the parameter ξ has influence both on the value and fuzziness of the lengths of the elaborated paths. Lower values of ξ result in shorter but more fuzzy paths. This enables various trade-offs depending the constraints of the task.

4.3. Linear regression with fuzzy data

In this example, we would like to present the application of extensional fuzzy numbers for the linear regression. This problem is common in the elaboration of fuzzy model in the Takagi-Sugeno-Kang neuro-fuzzy systems, where the consequences of rules are frequently elaborated with the linear regression.

Extensional fuzzy numbers can be applied to the maths formulae for crisp numbers without any modification of the algorithm. Linear regression for a set of points $\mathbb{X} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ can be expressed as a minimisation tasks for the loss function

$$L(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2, \quad (36)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & \mathbf{x}_1^T \\ 1 & \mathbf{x}_2^T \\ \vdots & \vdots \\ 1 & \mathbf{x}_n^T \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \quad (37)$$

Minimising (36) gives

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (38)$$

Let's apply the formula (38) to a set of 2D Gaussian numbers: $\mathbf{x}_1 = [(1, 0.1), (1.5, 0.2)]$, $\mathbf{x}_2 = [(2, 0.2), (4.5, 0.1)]$, $\mathbf{x}_3 = [(3, 0.3), (5.5, 0.2)]$, $\mathbf{x}_4 = [(4, 0.4), (8.2, 0.1)]$, and $\mathbf{x}_5 = [(5, 0.4), (9.5, 0.3)]$. With the extensional fuzzy numbers the result is $\mathbf{w}_1 = (1.97, 0.4)$ and $\mathbf{w}_2 = (-0.07, 0.4)$. The points and regression line are presented in Fig. 16. Fuzzy numbers are drawn as ellipses with semi-minor and semi-major axes representing the fuzzification in X and Y axes.

When operating with α -cut based arithmetic of fuzzy number (with 2000 α -cuts) the fuzzification of results elaborated with Eq. (38) exceed value 500, what makes the result uninterpretable. Eq. (38) can be applied for both for crisp numbers and extensional fuzzy numbers without need of any iterative numerical procedures.

5. Conclusions

The paper presents an application of extended fuzzy numbers (extensional fuzzy numbers) to the problem of ordering of fuzzy numbers. The extensional fuzzy numbers with operations and relational operations enable operations on fuzzy terms directly without any need for defuzzification, integration or any iterative procedures. The construction of extensional fuzzy numbers is simple and operations on extensional fuzzy numbers are computationally efficient. They avoid the fuzziness spread problem: the fuzziness of extensional fuzzy numbers does not increase with the number of operations.

This enables application of extensional fuzzy numbers in loops and iterations. Multiple types of extensional fuzzy numbers can be constructed with T-norms, which makes the number of types of extensional fuzzy numbers infinite. The application examples presented in the paper show that extensional fuzzy numbers can be successfully and efficiently applied in algorithms instead of real numbers in problems where uncertainty and imprecision are present without modifications of the algorithms. The C++ implementation of extensional fuzzy numbers is available from the free public GitHub repository: github.com/ksiminski/ksi-extensional-fuzzy-numbers

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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